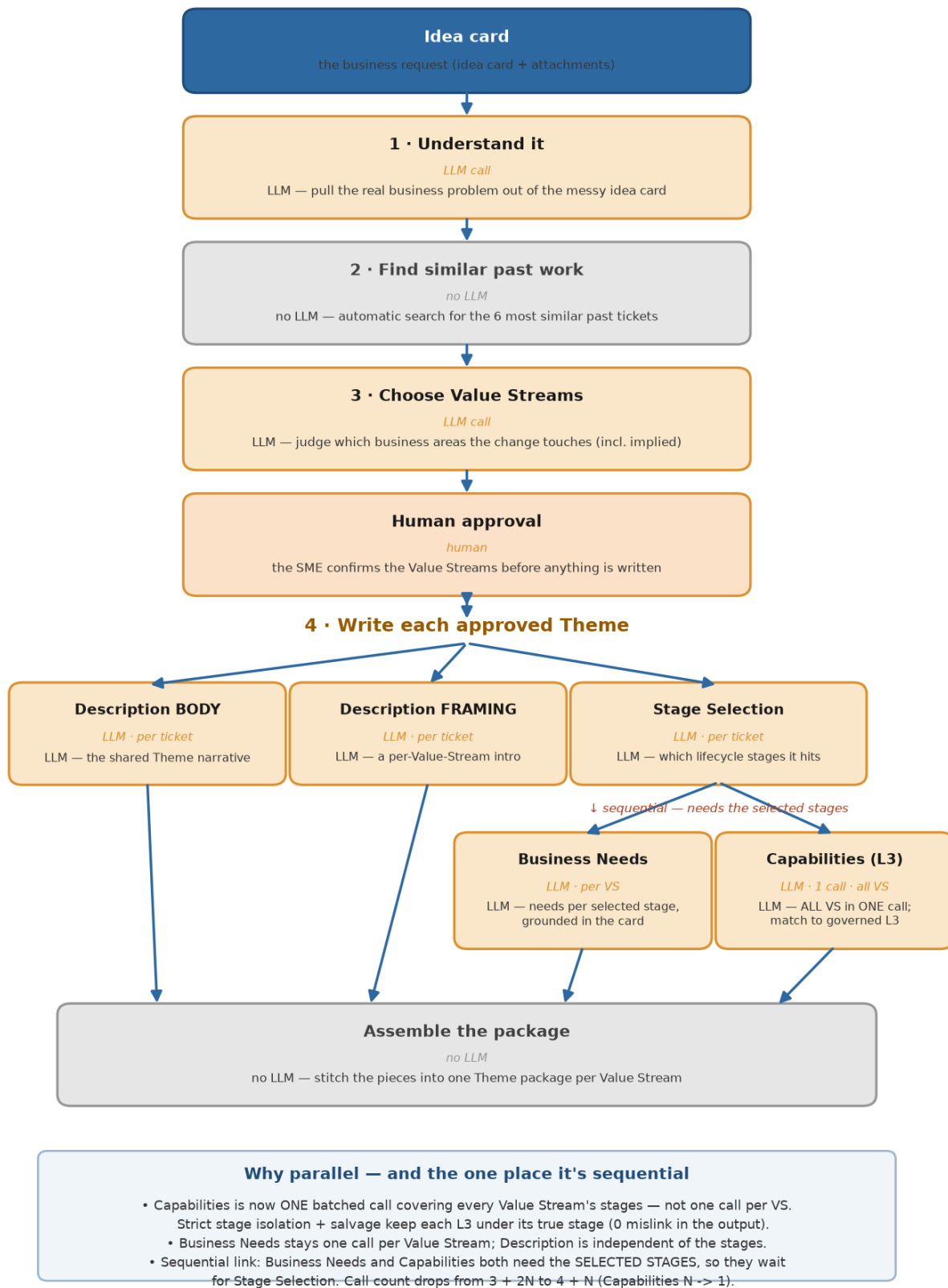


Generation flow — batching optimisations: cost & impact analysis

Two ways to cut the per-Value-Stream call count in theme generation, measured against the current per-VS architecture: **merged Capabilities** (all Value Streams' L3 in one call) and **batched Business Needs** (a few Value Streams per call). This is the cost and quality impact of each.

Generation flow (merged) — Capabilities in ONE call for all Value Streams



GPT-5-mini pricing throughout: **\$0.25 / 1M input, \$2.00 / 1M output.**

Baseline cost — current architecture (all per-VS), 10 Value Streams

Average tokens (the measured sample's idea cards sit below the 24k cap):

call	runs	input	output	cost
Condense	1	6,000	500	\$0.0025
Choose Value Streams	1	12,000	1,500	\$0.0060
Stage Selection	1	7,602	1,273	\$0.0044
Description BODY + FRAMING	2	10,304	1,706	\$0.0060
Business Needs	10	55,200	15,670	\$0.0451
Capabilities (L3)	10	58,470	6,990	\$0.0286
Total (average)	25 calls			\$0.093

24k worst case — every call carries the full 24k raw idea-card text:

call	runs	input	output	cost
Condense	1	24,000	500	\$0.0070
Choose Value Streams	1	30,000	1,500	\$0.0105
Stage Selection	1	26,600	1,273	\$0.0092
Description BODY + FRAMING	2	48,400	1,706	\$0.0155
Business Needs	10	245,200	15,670	\$0.0926
Capabilities (L3)	10	248,500	6,990	\$0.0761
Total (24k)	25 calls			\$0.211

So the baseline is \$0.093/ticket average, \$0.211 worst case (10 VS).

The changes and how they impacted quality

Merged Capabilities (L3) — all Value Streams in one call

metric	per-VS	merged
precision	0.53	0.43
recall	0.90	0.84
F1	0.67	0.57
mislink in output (after salvage)	0	0
hallucinated ids	0%	0%

Recall –6, strict precision –10 (much of which is plausible picks the ground truth didn't tag, not real error). Nothing hallucinated; mislink is 0 in the output either way. **Quality holds up well** — L3 is short *selection*, which batches cleanly.

Batched Business Needs — a few Value Streams per call

metric	per-VS	batched
faithfulness	0.895	0.715
hallucination	0.105	0.285
coverage	0.810	0.519

Faithfulness -18, coverage -29, hallucination +18. **Quality degrades materially.** Business Needs is long-form *writing*: the batched call emits **~920 output tokens per VS vs 1,567 per-VS** — each document is ~40% shorter, so it reflects fewer source facts and grounds fewer claims. The shorter output is the direct cause.

Cost comparison 1 — merged Capabilities (whole flow, 10 VS)

	baseline (per-VS)	merged Capabilities
LLM calls	25	16
cost — average	\$0.093	\$0.082
cost — 24k worst case	\$0.211	\$0.157

Capabilities collapses from 10 calls to 1: **9 fewer calls, -12% cost average, -26% at 24k** (the raw text is sent once instead of 10 times — the saving grows with idea-card size).

Cost comparison 2 — batched Business Needs (whole flow, 10 VS)

	baseline (per-VS)	batched Business Needs
LLM calls	25	20
cost — average	\$0.093	\$0.073
cost — 24k worst case	\$0.211	\$0.168

Business Needs goes from 10 calls to 5 (chunked): **5 fewer calls, -21% cost average, -20% at 24k.**

Verdict

change	calls saved	cost saved (avg / 24k)	quality
merged Capabilities	9	12% / 26%	small, acceptable trade
batched Business Needs	5	21% / 20%	materially worse (shorter, less-grounded docs)

Adopt merged Capabilities — it's the bigger call-count saving, the cost drop grows on large idea cards, and the quality trade is small (and nothing is hallucinated). **Keep Business Needs per-VS** — the saving is real, but it comes by making a prescriptive, architect-facing artifact ~40% shorter and notably less grounded, which is a bad trade.